1.8 Determining Solutions to Inequalities

Lesson Introduction

Benchmark	MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.		Learning Targets	• I can determine which value from a set of values make an inequality true. • I can determine if a value on a number line represents a solution set. • I can solve inequalities and represent solutions on a number line.
Benchmark Coherence	Building On			Working Toward
	MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.			MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.
Essential Questions	• How do you determine if a value makes an inequality true? • How do you determine the solution to an inequality? • How many solutions can an inequality have?			
Lesson Narrative	In this lesson, students work with equations and inequalities to find all possible solutions and compare the steps they take to solve each. Students begin by determining the possible solutions for given inequalities and equations and explain the strategies they used to find the solutions. Students continue to use similar strategies to solve equations and inequalities and graph inequalities on the number line.			
Component Summary	1.8.1 Warm-Up (~5 min.)	Explore forester career *(SE p. 47)*	1.8.4 Your Turn! (5-10 min.)	Solve two-step inequalities and graph *(SE p. 50)*
	1.8.2 Exploration (10-15 min.)	Explore possible solutions to inequalities *(SE p. 47)*	1.8.5 Collaboration (5-10 min.)	Match inequalities to solutions with a partner *(SE p. 50)*
	1.8.3 Guided Practice (15-20 min.)	Discuss and solve two-step inequalities *(SE p. 48)*	1.8.6 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 51)*
Resources	Glossary	Materials		Additional Preparation
	Key Terms: • Inequality Additional Terms: N/A	• Career Profile Video • Cards for 1.8.5 Collaboration matching activity • (Optional) Algebra tiles • (Optional) Paper		• 1.8.1 Warm-Up contains a video to accompany the question prompts.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not understand why they need to flip the inequality sign when dividing or multiplying by a negative number.
Students may not understand when to use a closed circle and when to use an open circle when graphing inequalities on a number line.</td><td>Students are still discovering the differences between equations and inequalities. For those who struggle, consider providing a graphic organizer or a visual prompt to help determine the differences at a glance.
Consider giving students a list of the properties that are used in 1.8.3 Guided Practice.
If time is an issue, consider assigning 1.8.4 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Clarify, Critique, Correct
Consider having partners do every other step when solving an inequality and correct their partner if they complete a step wrong or follow their partner if they do a step a different way but correctly.</td><td>MA.K12.MTR.1.1: Participation in Effortful Learning
Students will be participating in effortful learning, both individually and with others, as they analyze the relationship between the algebraic solution and the graphical representation of the solution. Students will help and support each other while attempting to learn new methods and approaches. Finally, students work together to match cards, which may require perseverance and asking questions.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider allowing students to record themselves instead of writing their reflections to help articulate their thinking. Encourage students to write out their work on a separate sheet of paper. Students may forget to keep the minus/negative sign after certain steps in the problem. Consider having students circle each term in the problem to help students visualize which signs belong with which terms. Consider allowing students to use algebra tiles to help kinesthetic learners with solving the inequalities.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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1.8.1 Warm-Up: Careers - Forester

Component Overview: *In this Warm-Up, students will explore the career of a forester. Show students videos and information about foresters. Students will discuss characteristics that would make them good at this job as well as other jobs.*

1.8.1 Warm-Up: Careers - Forester

Foresters have a wide range of duties, depending on who their employer is. Some primary duties of foresters are drawing up plans to regenerate forested lands, monitoring the progress of those lands, and supervising tree harvests. Another duty of a forester is devising plans to keep forests free from disease, harmful insects, and damaging wildfires.

1. Why do you think being an effective communicator and a team player is essential to the job?
   Answers will vary.

2. Describe a time when you had to be an effective communicator and team player.
   Answers will vary.

Play the Career Profile video on the forester for the entire class. After students watch the video (2:24 minutes), prompt them to reflect on the profession by answering the two questions independently.

Consider allowing students to write about what would happen at a job if an employee was neither an effective communicator nor a team player.

1.8.2 Exploration: Is It a Solution?

Component Overview: *In this Exploration, students will determine the multiple solutions for the given inequalities. Students will also predict the solutions for both inequalities and discuss how the inequalities are compared to each other. Consider providing sentence stems to help frame student discussion with their partner.*

1.8.2 Exploration: Is it a Solution?

A solution to an inequality is any value that, when substituted for the variable, makes the inequality statement true.

A list of possible solutions for the inequalities $2x - 5 > 13$ and $2x - 5 < 13$ is shown.

Possible Solution Values			
9.01	-4000	11.47	0
$115\frac{3}{4}$	-9	9	-6.25
8.375	10	38	$4\frac{1}{6}$

1. Determine whether each of the possible values is a solution to the given inequalities.

   a. Record the solutions in the table.

$2x - 5 > 13$	$2x - 5 < 13$	Not a Solution to Either
9.01 11.47 $115\frac{3}{4}$ 10 38	0 -4000 -9 -6.25 8.375 $4\frac{1}{6}$	9

   b. Explain the strategy you used to identify solutions.
   Answers will vary. I solved each inequality and chose all the numbers that were greater than 9 for the first box and all the numbers that were less than 9 for middle box. I substituted each number into the inequalities to see if they made it true.

Provide sentence starters to aid students who may struggle to explain their strategies in writing.

- *"I solved each inequality by first…then I…"*
- *"The strategy I used was…substitution/solving and checking…First I…"*

1.8.2 Exploration: Is It a Solution? (continued)

An inequality has infinite solutions and can be represented on a number line.

2. Discuss with your partner why an inequality has infinitely many solutions.
Either way the answers go on the number line, the numbers never end.

3. Predict the solution set for $2x - 5 > 13$.
$x > 9$

4. Predict the solution set for $2x - 5 < 13$.
$x < 9$

5. There are no values for x that satisfy both inequalities. Discuss with your partner why that is the case. Summarize your discussion.
The inequalities use the same numbers, but their signs are the reverse of each other and do not include an equal sign.

6. Neither inequality had a solution of 9. How could the inequalities $2x - 5 > 13$ and $2x - 5 < 13$ be rewritten so that 9 would be part of the solution set?
$2x - 5 \geq 13$
$2x - 5 \leq 13$

MA.K12.MTR.1.1: Participation in Effortful Learning
As you circulate, listen for students who are staying engaged and maintaining a positive mindset. Encourage students to help and support each other as they attempt to understand the solutions of an inequality.

1.8.3 Guided Practice: Solving Two-Step Inequalities

Component Overview: *Students justify each step when solving an inequality and an equation side by side. Students will discuss the similarities and differences between equations and inequalities. Students will also practice solving inequalities and graphing their solutions.*

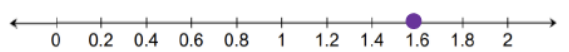
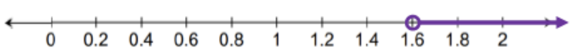
1.8.3 Guided Practice: Solving Two-Step Inequalities

Inequalities can be solved using similar strategies that are used for solving equations.

1. The work for solving the equation $17 = 5x + 9$ and the work for solving the inequality $17 < 5x + 9$ is shown.
   a. Complete the table with the justification of each step used for solving the equation and the inequality.
   Answers may vary.

Equation	Justification	Inequality
$17 = 5x + 9$	*Given*	$17 < 5x + 9$
$-9 \qquad -9$	*Subtract 9 from both sides*	$-9 \qquad -9$
$\frac{8}{5} = \frac{5x}{5}$	*Divide both sides by 5*	$\frac{8}{5} < \frac{5x}{5}$
$1.6 = x$	*Solution*	$1.6 < x$

The solution for the equation and the solution set for the inequality is shown.

$1.6 = x$	
$1.6 < x$	

   b. Discuss with your partner the similarities and differences between solving and graphing the solutions for equations and inequalities. Summarize your discussion.
   Answers will vary. Equations and inequalities are solved similarly until the last step if dividing or multiplying by a negative number. The answers are graph differently because the answer to the equation is one closed circle, while an inequality can be graph with an open or closed circle and a line drawn greater or less than the number.

Knowing which property is used should not be a barrier to student success on the Guided Practice. Students should be focused on the similarities between solving the inequality and solving the equation. Prompt students to share what they wonder and what they notice about the steps before completing the justifications. After students have had a chance to write down their responses, ask several students to share things they noticed and things they wondered about the comparison. Make connections between students' justifications and the properties of equality and inequality. A goal here is for students to see the similarities between the properties of equality and properties of inequality.

Consider allowing students to use a word wall or list of the justifications to refer to when determining the best one to use for the equation and matching inequality.

Students may need a graphic organizer to organize their thoughts before discussing their ideas with their partner. Consider providing students with a Venn diagram to visually compare the equations and inequalities.

1.8.3 Guided Practice: Solving Two-Step Inequalities (continued)

2. The work for solving the equation $13 = -5t - 7$ and the inequality $13 \geq -5t - 7$ is shown.
Answers may vary.

Equation	Justification	Inequality
$13 = -5t - 7$	*Given*	$13 \geq -5t - 7$
$+7 \qquad +7$	*Add 7 to both sides*	$+7 \qquad +7$
$20 = -5t$	*Divide both sides by -5*	$20 \geq -5t$
$\overline{-5} \quad \overline{-5}$		$\overline{-5} \quad \overline{-5}$
$-4 = t$	*Solution*	$-4 \leq t$

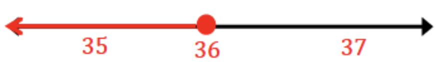
a. Complete the table with the justification of each step used for solving the equation and the inequality.

b. How did the final step of solving the inequality $13 \geq -5t - 7$ differ from the final step of solving the inequality $17 < 5x + 9$?
In the final step of $13 \geq -5t - 7$, the inequality sign needs to be flipped due to dividing by a negative number.

When the variable term in an inequality is negative, the inequality can be rewritten by multiplying or dividing the inequality by -1. In the resulting inequality, each term and the comparison symbol will change to its opposite.

3. Solve each inequality and graph the solution.

a. $2 - \frac{a}{6} \geq -4$
$a \leq 36$



b. $13 \geq -5t - 7$
$t \geq -4$

Some students may struggle to understand the inequality sign must be flipped to make the inequality true. Prompt students to consider why this is through questioning.
What do you notice about the inequality sign in the final step?
What would happen if the inequality sign didn't flip?
Would the solution of -4 still make the equation true?
When does the inequality symbol flip during solving? Why?
It may be helpful to write a simplified inequality such as $-x \leq 5$ and ask students to write a list of values that would be a solution. This will help students understand that the negative changes the sign of the solution and thus changes location on the number line.

Students may struggle with the steps for question 3a. Consider how to support students as they encounter $-\frac{a}{8}$. Students with a strong understanding of equivalence understand that $-\frac{a}{8} = \frac{-a}{8} = \frac{a}{-8}$. From a long-term student-understanding perspective, it would be better if your support utilizes equivalence. For some students, it may be best to begin with $-5x = \frac{-5x}{1} = \frac{5x}{-1}$, then to build to the equivalence needed for question 3a.

1.8.4 Your Turn!

***Component Overview**: In this Your Turn!, students will prove their knowledge of solving the inequalities and graphing the answers.*

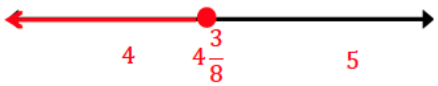
1.8.4 Your Turn!

Consider working with a small group of students who may need support during Your Turn!.

1. Solve each inequality and graph the solution.

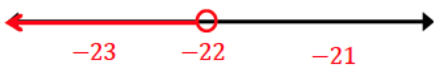
   a. $56 \geq 8x + 21$

   $x \leq 4\frac{3}{8}$



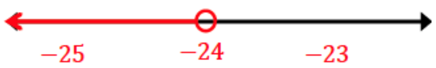
   b. $-2.7 - 0.5t > 8.3$

   $t < -22$



   c. $6 + \frac{3}{4}k < -12$

   $k < -24$



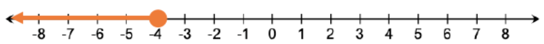
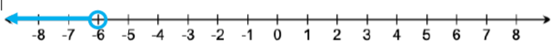
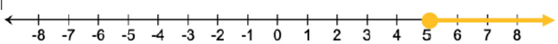
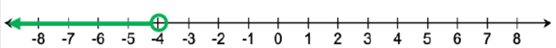
1.8.5 Collaboration: Matching Inequalities

***Component Overview**: In this Collaboration, students will work with a partner to solve inequalities. Students will be given cards with inequalities on them and need to match them to the given graphs.*

1.8.5 Collaboration: Matching Inequalities

Work with your partner to match each of the given inequalities to their graph of possible solutions.

1. Record your matches in the table.

Graph	Inequality
	$10 + 5x \leq -10$
	$9 < -\frac{x}{3} + 7$
	$-x + 11 \geq 6$
	$11 > -1 - 2x$
	$-4 + 0.2x \geq -3$
	$\frac{1}{2}x - 5 < -7$

2. What strategies did you use to match each inequality to its graph?
   Answers will vary. I solved all the inequalities first and then matched the answers. I substituted the solutions into each inequality to find which one was true.

MA.K12.MLR.3.1: Clarify, Critique, Correct
Consider having partners do every other step when solving an inequality and correct their partner if they complete a step wrong or follow their partner if they do a step a different way but correctly.

1.8.6 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

1.8.6 Wrap-Up: Cool Down

1. Explain a way that you can determine if a solution is correct for a given inequality.

   Answers will vary.

Consider allowing students to record themselves instead of writing their response to help articulate their thinking.

2. How do you know if the circle should be open or closed when graphing an inequality?

   Answers will vary. If the comparison symbol used is $<$ or $>$, an open circle should be used because the value is not included in the solution set. If the comparison symbol used is $\leq$ or $\geq$, a closed circle should be used because the value is included in the solution set.

Consider pairing students to discuss their responses to see if peers can answer any of their questions.